

Tide retrospective: affine-observable κ_3 for the anharmonic Gibbs (G2)

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Abstract

This retrospective records a 100-line trilinear strict-improvement of Tide 9. Tide 9 proved $t^2 \cdot \kappa_3(x, x, x) \rightarrow -\alpha/\lambda^3$ for the 1D anharmonic Gibbs measure with potential $L = (\lambda/2)x^2 + (\alpha/6)x^3 + (\gamma/24)x^4$. This tide upgrades the result to arbitrary id-multiples $(b\,x, a\,x, c\,x)$, giving $t^2 \cdot \kappa_3(b\,x, a\,x, c\,x) \rightarrow -(abc)\alpha/\lambda^3$, via the static factorisation $\kappa_3(b\,x, a\,x, c\,x) = (abc) \cdot \kappa_3(x, x, x)$. The factorisation is hypothesis-free — it follows by trilinearity of κ_3 once each of the seven Gibbs expectations in `Threepoint.kappa3`'s definition has its scalar factor peeled off via I2's just-landed `gibbsExpectation_smul`.¹ The asymptotic is then `Tendsto.const_mul (abc)` on top of Tide 9. Zero `sorry`, single session, build green on the first attempt; the GPT-5.5 Pro deliberation agreed with the Claude draft and flagged a deferred shifted-affine extension.

1 Setting

Tide 9 (anharmonic- κ_3 , 6 May) proved

$$t^2 \cdot \kappa_3(\text{volume}, L_{\lambda\alpha\gamma}, x, t, x, x) \rightarrow -\alpha/\lambda^3 \quad \text{as } t \rightarrow \infty$$

for $L_{\lambda\alpha\gamma}(x) = (\lambda/2)x^2 + (\alpha/6)x^3 + (\gamma/24)x^4$, under the discriminant condition $\alpha^2 < 3\lambda\gamma$.² The slot tuple (id, id, id) specialises κ_3 's general (A, φ, B) signature to all-identity observables. Tide 9 was the first cross-repo bridge between the abstract `kappa3` machinery in `threepoint` and the concrete anharmonic asymptotics in `laplace`; its retrospective named the affine-observable form as the natural multilinear strict-improvement and estimated ~ 40 lines.

I2 (gibbsCov-algebra, 7 May) landed an unconditional Gibbs-expectation / Gibbs-covariance algebra:³ Of these, the only load-bearing one for this tide is `gibbsExpectation_smul` : $\langle c\varphi \rangle_t = c \cdot \langle \varphi \rangle_t$, which holds unconditionally (when the partition is zero, both sides are zero). The trilinearity factorisation reduces to seven applications of this identity plus a `ring` step to collect the scalar factor abc ; no `gibbsCov` algebra is needed.

This tide was picked at 2026-05-07T20:55Z from candidate G2 in the v3 candidates survey,⁴ after a parallel session claimed C1 (the harmonic-side twin) at 20:54Z. Together with the in-progress C1, the two land complete affine-observable κ_3 instances across both harmonic and anharmonic sides on the same day.

¹I2 (gibbsCov-algebra) landed 2026-05-07. Branch: `tide/gibbscov-algebra`.

²Discriminant condition guarantees the sextic Wick contractions in $\langle x^4 \rangle_t$ stay positive, so the asymptotic of the covariance $\langle x^2, x \rangle$ has the sign Tide 9 records.

³Eleven lemmas mirrored across `Laplace/Gibbs.lean` and `Laplace/Multi/Basic.lean`: `gibbsExpectation_{smul, add, zero, const}` and `gibbsCov_{symm, smul, const, add, zero}`.

⁴`projects/automation/log/2026-05-07-v3-tide-candidates-survey.md`.

2 The thing formalised

Two new public theorems in `Laplace/OneD/AnharmonicKappa3.lean`.

Theorem 1 (potential-agnostic factorisation). For any $L : \mathbb{R} \rightarrow \mathbb{R}$, $t \in \mathbb{R}$, and scalars $a, b, c \in \mathbb{R}$,

$$\kappa_3(\text{volume}, L, b \cdot x, t, a \cdot x, c \cdot x) = (a \cdot b \cdot c) \cdot \kappa_3(\text{volume}, L, x, t, x, x).$$

Lean signature:

```
theorem kappa3.affine_id_id_id_eq
  (L : ℝ → ℝ) (t : ℝ) (a b c : ℝ) :
  Threepoint.kappa3 (volume) L
    (fun x => b * x) t (fun x => a * x) (fun x => c * x)
  = (a * b * c) * Threepoint.kappa3 (volume) L
    (fun x => x) t (fun x => x) (fun x => x)
```

The factorisation has no analytic preconditions: κ_3 's seven ratio-of-integrals reduce, at $h = 0$, to seven `Laplace.gibbsExpectation` ratios via the existing `threepoint.gibbsExp_volume_zero_eq` bridge, and `gibbsExpectation_smul` is unconditional.

Theorem 2 (anharmonic asymptotic). For $L_{\lambda\alpha\gamma}$ with $0 < \lambda$, $0 < \gamma$, $\alpha^2 < 3\lambda\gamma$, and any $a, b, c \in \mathbb{R}$,

$$t^2 \cdot \kappa_3(\text{volume}, L_{\lambda\alpha\gamma}, b \cdot x, t, a \cdot x, c \cdot x) \rightarrow -(a \cdot b \cdot c) \cdot \alpha / \lambda^3 \quad \text{as } t \rightarrow \infty.$$

Lean signature:

```
theorem kappa3.anharmonic_affine_asymptotic
  {lam alpha gamma : ℝ}
  (hlam : 0 < lam) (hgamma : 0 < gamma) (hdisc : alpha^2 < 3 * lam * gamma)
  (a b c : ℝ) :
  Filter.Tendsto
    (fun t : ℝ => t^2 * Threepoint.kappa3 (volume)
      (anharmonicPotential lam alpha gamma)
      (fun x => b * x) t (fun x => a * x) (fun x => c * x))
    Filter.atTop
    (nhds (-(a * b * c) * alpha / lam^3))
```

The hypotheses are exactly those of Tide 9: trilinearity of κ_3 in the three observable slots costs nothing analytically.

3 Proof strategy

Theorem 1. The key observation is that `Threepoint.kappa3` unfolds to seven `gibbsExp` terms,⁵ all evaluated at $h = 0$. The bridging lemma `threepoint.gibbsExp_volume_zero_eq` (already in `laplace`, from Tide 9) routes each one through to `Laplace.gibbsExpectation`, after which the seven integrand λ -binders carry products like $(\text{fun } x \mapsto ax)w \cdot (\text{fun } x \mapsto bx)w$ that are not

⁵Specifically: $\langle \varphi AB \rangle$, $\langle \varphi A \rangle \langle B \rangle$, $\langle \varphi B \rangle \langle A \rangle$, $\langle AB \rangle \langle \varphi \rangle$, and $\langle \varphi \rangle \langle A \rangle \langle B \rangle$ — a polynomial of total degree at most three in the seven scalar values.

yet syntactically in the canonical scaled-monomial form. A `funext; ring` cleanup⁶ forces them into the canonical shapes. `gibbsExpectation.smul` then peels each scalar factor out of its Gibbs expectation, leaving a pure ring identity in $(a, b, c) \in \mathbb{R}^3$ and the four scalar values $\langle w^3 \rangle$, $\langle w^2 \rangle$, $\langle w \rangle$, $\langle w \rangle^3$ (which appear cubically in the standard polarisation $\langle w^3 \rangle - 3\langle w^2 \rangle \langle w \rangle + 2\langle w \rangle^3$). `ring` closes.

Theorem 2. The new `Tendsto` is just abc times Tide 9’s: $t^2 \cdot \kappa_3(bx, ax, cx) = abc \cdot t^2 \cdot \kappa_3(x, x, x)$ pointwise. So the proof is `Tendsto.const_mul (abc)` applied to the Tide 9 conclusion, combined with `Tendsto.congr’` rewriting the function under the integral via Theorem 1. The constant $-(abc) \alpha / \lambda^3$ is normalised against $abc \cdot (-\alpha / \lambda^3)$ by a one-line `ring`.

4 Roadblocks and resolutions

There were no roadblocks. The proof landed green on the first `lake build`, in 4.7s after a warm Mathlib cache. This is unusual but reasonable for a tide whose math is pure trilinearity over a closed-form cumulant; the load-bearing analytic content is fully imported from Tide 9.

The deliberation step ran in parallel with the formalisation in this session, on the assumption (correct, as it turned out) that the candidate was uncontroversial. GPT-5.5 Pro’s response, when it returned 4 minutes after the proof had already landed, confirmed Candidate A as the right patch-sized theorem and noted three small points worth recording:

1. *The asymptotic constant.* GPT confirmed the sign $-(abc) \alpha / \lambda^3$ (no factor or sign error). Suggested stating the limit as `((a * b * c) * (-alpha / lam^3))` for Lean ergonomics; the proof uses `-(a * b * c) * alpha / lam^3` and bridges the two forms by a single `ring` step. Both work; one wonders if Mathlib will eventually grow a normal-form simp set for these.
2. *Algebra path.* “You likely don’t need `gibbsCov` for this patch; direct `gibbsExpectation.smul + gibbsExpectation.const` is enough.” Exactly the route taken; the I2 `gibbsCov` algebra is not used here. (It is used in C1, the harmonic-side twin.)
3. *Missed candidate (deferred).* “The affine strengthening is also true: $\kappa_3(bx+d, ax+e, cx+f) = abc \cdot \kappa_3(x, x, x)$, because `kappa3` kills constants in each slot ($\mathbb{E}[1] = 1$ makes the cancellations exact). But I would defer it unless you need shifted observables now.” Recorded as a follow-up; not picked up here.

5 What was Mathlib, what was new

Mathlib. `Tendsto.const_mul` (constant multiple of a limit), `Tendsto.congr’` (rewriting the function under a limit), and `filter_upwards` (eventual quantification). All appear in dozens of laplace files; nothing new from Mathlib.

Laplace seabed (existing). Three load-bearing imports:⁷ Tide 9’s headline asymptotic, Tide 9’s $h = 0$ bridge to `Laplace.gibbsExpectation`, and I2’s unconditional scalar-out-of-Gibbs-expectation lemma.

⁶Seven such have `h_X : (fun w => ...) = (fun w => K * w^n)` steps: $abc \cdot w^3$, $ab \cdot w^2$, $ac \cdot w^2$, $bc \cdot w^2$, plus the three aw , bw , cw singletons, and the four RHS-side identity-only forms w^3 and w^2 .

⁷Tide 9’s `kappa3.anharmonic.id.id.id.asymptotic` (the headline), its private bridge `threepoint.gibbsExp.volume.zero.eq`, and I2’s `Laplace.gibbsExpectation.smul`.

New. Two public theorems and seven `funext; ring`-style integrand-cleanup `have`-blocks inside the proof of Theorem 1. Nothing new at the lemma library level. The seven `funexts` are mechanical bookkeeping; a future tide could fold them into a small `simp` set “`push_smul_under_lambda`” but at one use it’s not yet cost-positive.

6 Lessons

Trilinearity tides land in one shot when the algebra is already there. I2 paid for itself the moment a strict-improvement of this shape was on the board. The factorisation theorem is essentially the I2 `gibbsExpectation_smul` lemma applied seven times under a `ring`-closure; if I2 hadn’t been merged we’d have unfolded each expectation manually. The lesson generalises to C1 (harmonic-side twin) and is implicit in the I2 retrospective’s follow-up list, which flagged C1 and G2 specifically.

Parallel deliberation + formalisation is fine when the deliberation outcome is predictable. Auto mode here meant launching the GPT consult in the background and writing the Lean code in the foreground, without waiting. The risk is wasted work if GPT had named a different candidate; the actual outcome (GPT’s vote agreed with Claude’s) suggests this risk is acceptable for trilinearity-shaped tides where the candidate is essentially forced. Less acceptable for hypothesis-class deliberations or when the proof technique is genuinely contested. Don’t generalise this to all tides.

The “`funext; ring cleanup`” pattern is the hidden cost of the seven-term `kappa3` unfolding. Both Tide 9’s `kappa3_id_id_id_unfold` and this tide’s `kappa3_affine_id_id_id_eq` carry the same boilerplate: the seven Gibbs-expectation integrands need their `lambdas` reduced to canonical scaled-monomial form before `gibbsExpectation_smul` can fire on them. A second use isn’t yet a refactor trigger, but if a third `kappa3` unfolding tide lands (e.g. for $\langle x^k \rangle$ slots, or for the cross-cumulant $\kappa_3(\varphi, A, B)$ of the patterning paper), the pattern is well-developed enough to extract.

7 Follow-ups

- *Shifted-affine strengthening* (GPT’s missed candidate). $\kappa_3(bx + d, ax + e, cx + f) = abc \cdot \kappa_3(x, x, x)$ via `gibbsExpectation_const`-style cancellations ($\mathbb{E}[1] = 1$). ~50 extra lines on top of this tide. Defer until a second consumer for shifted observables materialises.
- *C1 (harmonic-side twin), in flight.* The harmonic case $\kappa_3(bx, ax, cx) = abc \cdot \kappa_3(x, x, x) = 0$ follows by the same trilinearity argument, on top of Tide 5’s harmonic-vanishing theorem.⁸ Claimed by a parallel session at 20:54Z; should land within hours of this retrospective. Together the two tides close the affine-observable κ_3 pair for the 1D potentials currently in `laplace`.
- *Generic multilinear factorisation (Candidate C, deferred).* $\kappa_3(\mu, L, aA_0, t, b\varphi_0, cB_0) = abc \cdot \kappa_3(\mu, L, A_0, t, \varphi_0, B_0)$ for arbitrary A_0, φ_0, B_0 . Subsumes Theorem 1 as the (id, id, id) instance. GPT noted “only worth exporting if it is essentially free after the $h = 0$ rewrite”; worth checking against C1’s proof to see whether a unified C-shaped helper would shorten both files. If yes, a small refactor tide.

⁸`Threepoint.kappa3_harmonic_id_id_id_eq_zero`.

- Promote “*push_smul_under_lambda simp set*” on third use. The seven-`funext` pattern (this tide and Tide 9) is a candidate for automation but not yet cost-positive. Third use triggers.
- `\leanref-tag` the *threepoint paper*, pass 3. This tide is a corollary of the threepoint paper’s multilinearity-of- κ_3 remark.⁹ Once the next process-tide tagging pass is run (currently candidate K2), tag the multilinearity claim with `\leanref` pinned to this commit.

⁹Roughly the slot-by-slot linearity of cumulants used implicitly in the patterning subtractions.